

Compiler Design Lab

Assignment 01: Lexical Analyzer in C++

[Deadline: 01/07/25; Time: 11:59 PM]

Develop a C++ program that reads a .c source file and identifies and classifies tokens into the following categories:

- **Keywords** (e.g., int, return)
- **Identifiers** (variable and function names)
- **Constants** (numeric literals)
- **Operators** (e.g., +, -, *, /, ==)

1. **Input:** A .c file (e.g., input.c) containing valid C code.
2. **Output:** A clear classification of all tokens found, printed to the console or written to an output file. Format example:

```
Keyword: int, float  
Identifier: main, num1  
Operator: =, +  
Constant: 55, 8659
```

Assignment 2: Parentheses Balance Checker in C++

[Deadline: 08/07/25; Time: 11:59 PM]

Write a C++ program that checks whether **all types of parentheses** in a given C source file (.c) are **balanced**.

However, parentheses that appear **inside string literals** (e.g., " ") or **comments** (both single-line // and multi-line /* ... */) **must be ignored**.